IEEE
AI Team : Task 4

Plotting
Basics in Python

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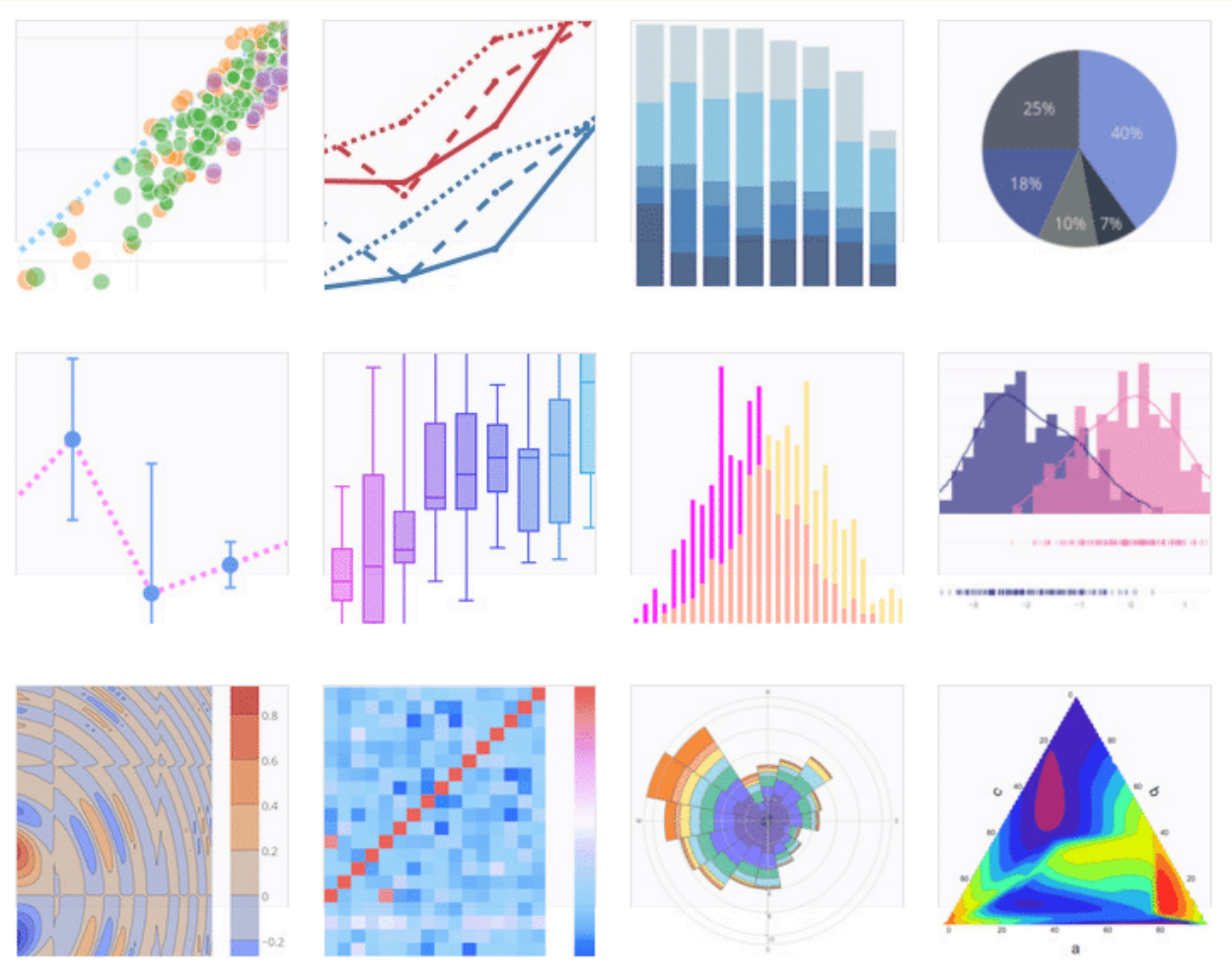


INTRODUCTION

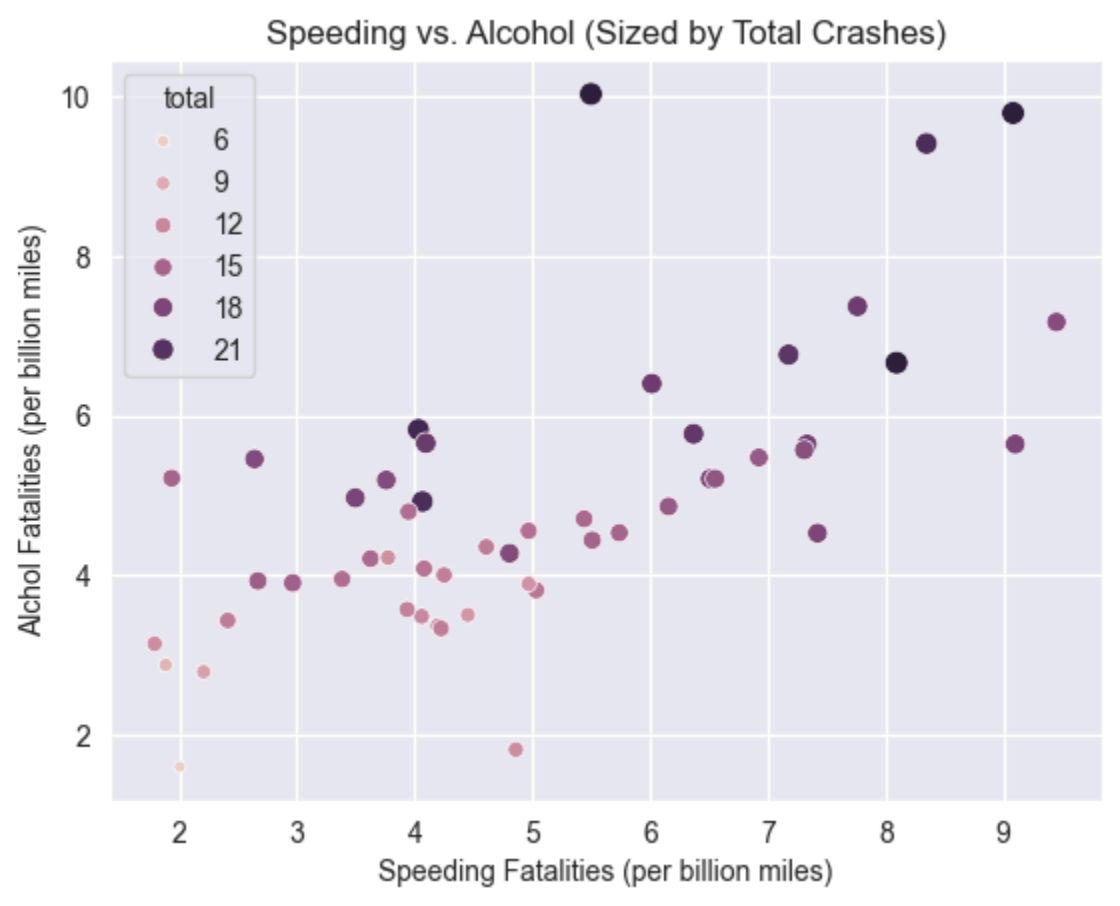
Plotting is an essential part of Interpreted languages in General and Python specifically. Known for its state of the art plotting libraries, it is the go to language for classic ML and data science.

Seabron and Matplot

Both are classics in the realm of visualizing data, I used both draw insights and have a better grasp of the car_crashes data set in Seaborn standard library and here are the findings.

Speeding and alcohol

First we explored the relation between speeding and alcohol in terms of over all number of crashes (per billion miles) using a scatter plot for visual clarity.

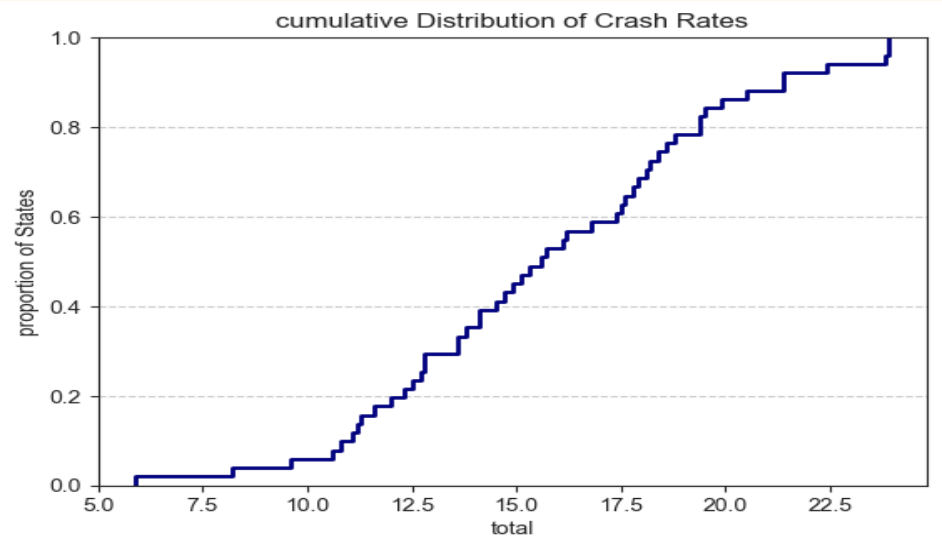


We first see a diagonal cluster, it indicates that the states with a high rate of speeding also tend to have a high rate of alcohol related fatal crashes. If a state has a high speeding number but a cool-color looking dot it indicates that speeding/drinking may not be actually the leading cause for fatal crashes in a given state.

These stats can help law makers in a given state to determine if they should loosen or tighten up speeding laws for example.

Rogue prediction

We will now look at an approximation of what the probability of a crash occuring would look like for the states using an ECDF plot.

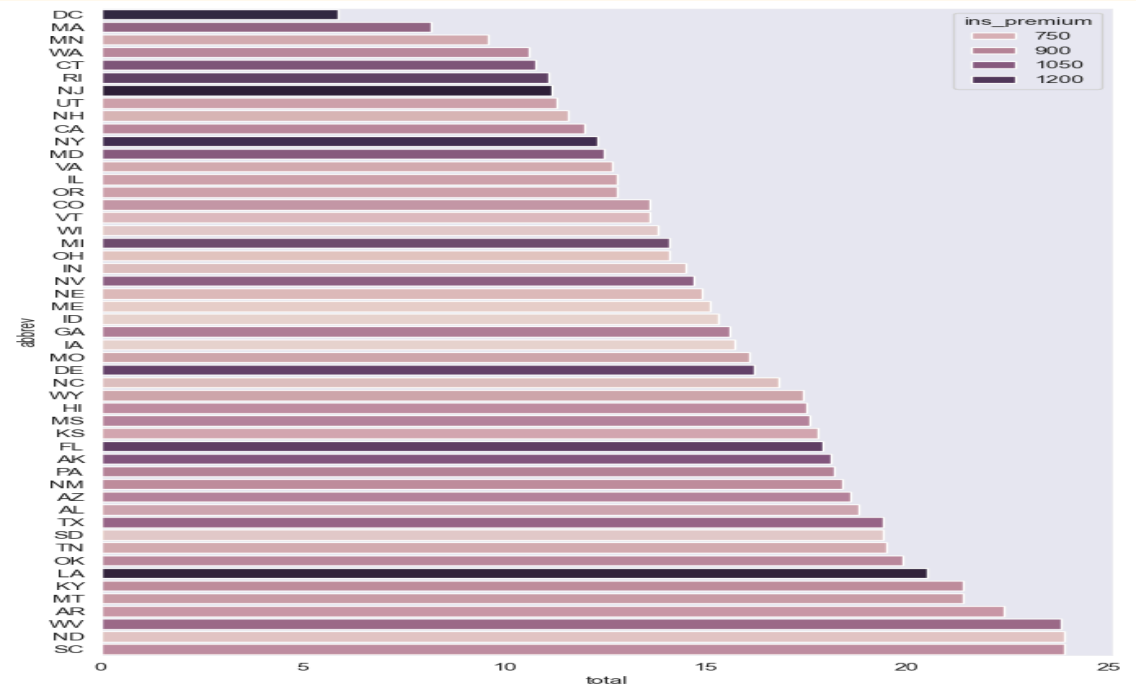


For each value on the x axis it shows the number of crashes per billion mile, the value of y axis shows what percent of states would have this number of crashes or lower.

As we can see about 60 percent of states have atleast 17.6 crashs, with a huge spike to 20.0 at around 80 percent, This indicates that most of the states need some refinement in their road laws.

States and Insurance

We will now explore the relationship between the total number of crashes for each indivisual state and the insurance avg of that state.



As we can see that states with highest number of crashes per billion miles seemed to have a relatively average or below average insurance pay.

This could suggest that drivers may tend to be more reckless at driving when knowing that the insurance companies would cover their losses which spikes the total number of crashes massively.

THE REST OF MY FINIDINGS AND GRAPHS ARE IN THE DATA_TASK.IPYNB FILE.

Difference between Matplot and Seaborn and when to use each

In essence Matplot is the engine while Seaborn is a high-end kit built on top of it, when using Seaborn you are by default using Matplot but with an abstraction layer.

This abstraction layer serves the purpose of finishing the most common task in data visualisations in a really well-made themes.

Concise, as it takes 1 line perform most common plots, and its optomized for Panda's frames. So in essence it allows you to focus on **what** the data is showing rather than **how** to show the data.

Matplot on the other hands gives you all the tools you would ever need but you need to tell it **everything** to show the data in a presentable way so it tends to be verbose.

It allows you to build custom charts that otherwise wouldnt be possible using Seaborn only.

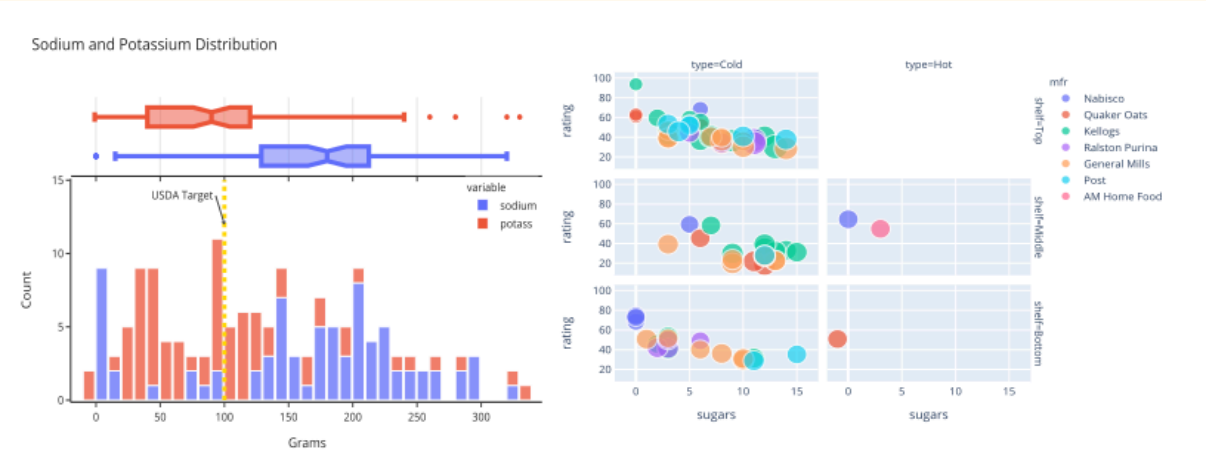
Matplot can also be used in dashboard apps and gui embedding without too much overhead, and its also used for production level papers that needs refinement of every pixel pretty much.

Ploty

If Matplot is the engine, Seaborn is the high-kit then Ploty is the touch screen dashboard.

Rather than only building static images, it allows you to build interactive, web based visualisations that you can hover and toggle data on and off, creating an entire story by a single graph.

Ploty charts are actually built using javascript, so its optimizd for web-applications and and online reports.

Plotly is the foundation for **Dash**, a framework used to build full-scale analytical web applications without needing to know HTML or CSS.

Below is an example of what a Ploty graph could look like.